Design Thinking Project: Modelling Uncertainty

Megan, Eddy, Hoe Teng, Zeyu, Edrea

Introduction/ Hypothesis

With the growing global population, the food security and demand for sufficient food has become increasingly urgent. Furthermore, climate change has significantly altered crop growth conditions, posing challenges to agricultural production, threatening food security around the globe. Thus, this report hypothesises that *crop yield* can be predicted by the *agricultural conditions* for each type of crop to identify and recommend the best crop for optimal yield under current climate conditions.

Problem Statement

HMW use Multiple Linear Regression to recommend crop types to farmers based on agricultural conditions to maximise yield?

1 Methodology

1.1 Dependent/Response Variable

The dependent variable is **Crop Yield per unit area** *(hectogram per hectare)* for each crop.

1.1.1 Crop Type

We chose to use a range of staple foods for our scope of recommendation, as they supply a large proportion of a person's energy and nutritional needs, which is critical for food security.

- **Cassava**
- **Corn**
- **Potato**
- **Rice**
- **Soybean**
- **Wheat**

1.2 Independent/Predictor Variables

- [**Average precipitation per year**] [1] *(mm/year)*: Provides water for crops to grow.
- [**Average Temperature per year**] [2] *(°C)*: Different crop types have different optimum temperatures for growth.
- [**Pesticides used**] [3] *(kg/ha)*: Protects crops from harmful pests and weeds that absorb nutrients away from them, increasing crop yield.
- [**PM 2.5**] [4] *(ng/m^3)*: High levels of PM 2.5 can negatively affect crops by reducing the sunlight reaching them, as the particles deposit on the leaves, decreasing crop yield.
- **Machinery per Agricultural Area** [5]: More machinery and technology should improve efficiency in preparation, irrigation, harvesting, etc., thus increasing crop yield. The unit area is 1000 hectares.

[1] https://ourworldindata.org/grapher/average-precipitation-per-year?tab=table#explore-the-data
[2] https://www.kaggle.com/datasets/palinatx/mean-temperature-for-countries-by-year-2014-2022?resource=download
[3] https://ourworldindata.org/grapher/pesticide-use-per-hectare-of-cropland?tab=table&time=1990..latest
[4] https://data.worldbank.org/indicator/EN.ATM.PM25.MC.M3?view=chart
[5] https://ourworldindata.org/grapher/machinery-per-agricultural-land?tab=table

2 Fine Tuning/ Iteration of data

To iterate and improve our model's accuracy, we decided to refine our data preparation with factors below:

2.1 Collinearity between the variables

To ensure the model is the most accurate, the predictor variables must be **independent** of each other as we want to avoid multicollinearity. Preliminary, we used correlation function in Excel to find any existing relationship.

	PM2.5 (µg/m^3)	*Annual Precipitation (mm)*	*Annual Mean Temperature (°C)*	*Pesticides Use per area of cropland (kg/ha)*	*Machinery*
PM2.5 (µg/m^3)	1				
Annual Precipitation (mm)	-0.144116381	1			
Annual Mean Temperature (°C)	0.437593448	0.208692036	1		
Pesticides Use per area of cropland (kg/ha)	-0.257347625	0.182462129	-0.019152139	1	
Machinery	-0.243451933	-0.017162331	-0.415266757	0.304645	1

Figure 1: Collinearity of independent variables

The highest collinearity is found between **Temperature** and **PM 2.5**, **Temperature** and **Machinery** (0.438 and -0.415 respectively). Even though Temperature is moderately collinear to 2 other variables, it does not make sense to remove temperature as temperature is a critical factor in determining crop yield. It directly influences plant growth and photosynthesis rates. Furthermore, logically speaking, it does not make sense for high surrounding temperature to increase in machinery use, thus, we preliminary conclude that the collinearity is purely coincidental.

2.2 Linearity relationship between each predictor and response variables

Next, we tried plotting the different crops' yields against the 5 predictor variables and calculated their **R-square values, individually**. This is to find whether each predictor variable is linearly relevant to the yield of each crop. Higher R-squared value would mean stronger linear relationship with the yield of each crop, whereas lower values suggest weaker linear relationship.

R-squared/Yield	**Cassava**	**Corn**	**Potato**	**Rice**	**Soybean**	**Wheat**	**Average R-squared**
PM2.5	0.0035	0.1018	0.1023	0.0673	0.1577	0.0995	0.0887
Annual Precipitation	0.0163	0.0219	0.0314	0.0054	0.0139	0.0033	0.0154
Annual Mean Temperature	0.0046	0.2959	0.1313	0.1584	0.1185	0.1878	0.1494
Pesticides per use area	0.0046	0.1123	0.1358	0.1139	0.1436	0.0742	0.0974
Machinery per use area	0.0273	0.1044	0.1320	0.0886	0.0760	0.1520	0.0967

Annual Precipitation has the lowest average R-squared value/weakest linear relationship with crop yield at 0.0154, probably due to the non-linear nature of its effect on crops. While precipitation is essential for crop growth, excessive or insufficient rainfall can negatively impact yield, and the optimal amount varies by crop type. However, we still decide to keep it as a predictor due to its paramount importance in plant growth. Although precipitation has weak linear relationship, including it helps account for variability in yield caused by water availability.

2.3 Graphs to display correlation between predictor and response variables

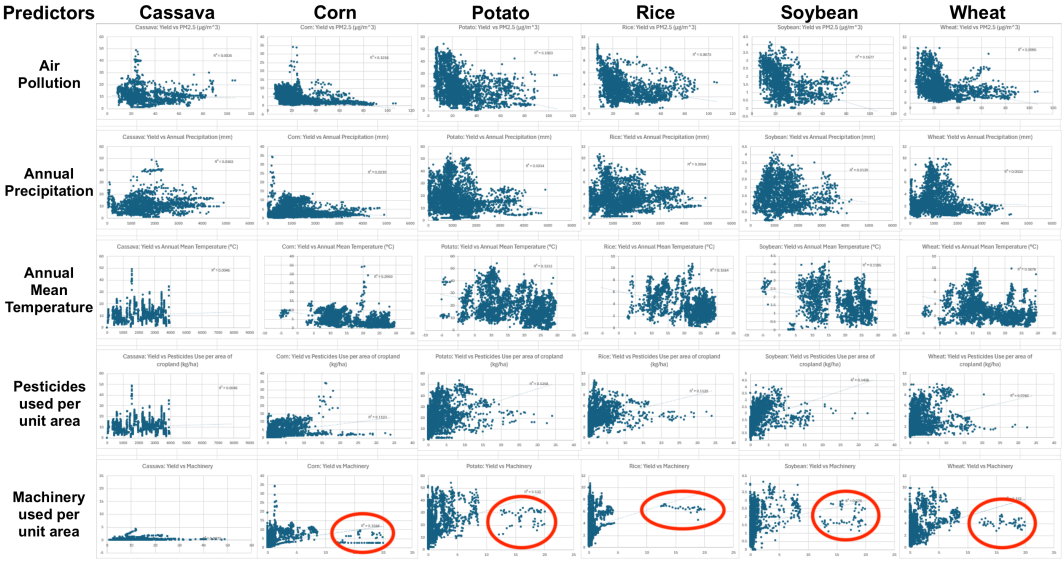


Figure 2: Graphs of each crops' yields against each predictors

The graphs in the bottom row exhibit a **lack of correlation** between the amount of machinery used and crop yield. While machinery usage is concentrated around a specific value, with only a few outliers, the resulting crop yields vary significantly across a wide range around the mean, resulting in its R-squared value being higher than some other variables despite its weaker linear relationship. This indicates that machinery usage alone does not appear to have a significant impact on crop yield, thus, we decided to **drop Machinery from the predictor variables**.

3 Model Development

3.1 Regression Equation

$$Y_i = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \beta_4 X_4$$

Where:

- Y_i: Dependent variable (*crop yield,* $\imath$*= crop types*)
- X_1, X_2, X_3, X_4: Independent variables; Average precipitation per year, Average Temperature per year, Pesticides used, PM 2.5 and Machinery per Agricultural Area, respectively
- $\beta_1, \beta_2, \beta_3, \beta_4$: Coefficients of independent variables
- β_0: Intercept

3.2 Model Summary

The coefficients and their statistical significance are summarised in the **appendix below on pg 6**

3.3 Model Analysis

3.3.1 Predictors Insights

- **PM2.5:**
  - All coefficients are negative, implying that more PM2.5 will decrease crop yield.
  - PM2.5 is statistically significant for most crops with small p-values. However, it is marginally significant for wheat $p = 0.062$, (in the gray area).

- **Annual Precipitation:**
  - Most coefficients are negative, implying that excessive precipitation negatively impacts yield, other than Cassava
  - Low p-values suggest that precipitation is a significant predictor for all crops.
- **Annual Mean Temperature:**
  - Most coefficients are negative, implying that higher temperatures tend to reduce crop yields, except for cassava, which shows a positive response.
  - Low p-values indicate that temperature is a significant predictor for all crops.
- **Pesticides Use per Area:**
  - All coefficients are positive, implying that more pesticide use will increase crop yield.
  - Low p-values suggest that pesticide usage is a significant predictor for all crops.

3.3.2 Response Insights

Based on the magnitude & sign of coefficients:

- **Cassava:** PM2.5 has a small negative impact, while temperature has a positive impact. Pesticides have a moderate positive effect.
- **Corn:** PM2.5 and precipitation negatively affect yield to a small extent, while pesticides significantly improve yield. More sensitive to temperature changes, doing well in lower temperatures.
- **Potato:** Strongest positive response to pesticide use among all crops. More sensitive to temperature changes as well, doing well in lower temperatures.
- **Rice:** Smaller coefficients for PM2.5, precipitation, and pesticides compared to other crops, indicating that Rice is less sensitive than other crops.
- **Soybean:** Smaller magnitude of coefficients for almost all predictors, but pesticides have a relatively small & positive impact.
- **Wheat:** PM2.5 is not a significant predictor. Stronger negative impacts from temperature and precipitation, with moderate improvement from pesticides.

3.3.3 Recommendations

- In general, PM2.5 suggests a lower crop yield; Annual precipitation had very low impact on crop yield; Temperature has varying impacts; Pesticide usage suggests higher crop yield.
- Reduce PM2.5 levels by having air quality control and optimise use of pesticides for all crops.
- Improve temperature control as crops have different optimal temperature to grow at.

4 Model Evaluation

4.1 Overall accuracy of Model

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}$$

We decided to use **Root Mean Squared Error** *(RMSE)* to assess the model's accuracy. Initially, we used **Mean Squared Error** *(MSE)*, however, we found that **MSE** for Cassava and Potato were significantly high. This was attributed to their higher yield values, which amplified prediction errors when squared. To address this issue, we switched to using **RMSE**, which retains the ability to penalise larger errors without disproportionately amplifying them based on yield magnitude. This shift allows for a more fair evaluation of model performance across different crops.

Crop Type	MSE(hg^2/ha^2)	RMSE(hg/ha)	Mean yield(hg/ha)	Ratio of RMSE to Mean Yield
Wheat	2.866	1.693	2.859	0.592
Soybean	0.456	0.676	1.578	0.428
Rice	2.511	1.585	3.436	0.461
Potato	75.420	8.684	18.204	0.477
Corn	6.485	2.547	3.660	0.696
Cassava	38.154	6.177	10.760	0.574

Overall, the **RMSE** are fairly large in comparison to the mean yield for the crops, which shows that the model is **not exceptionally accurate**. This is likely due to the fact that there are other qualitative independent variables (e.g. skill level of farmers, farmers' familiarity with the crops) which affect the dependent variable that were not taken into account. In addition, the factors, such as temperature as shown earlier, are not fully linearly related to the crop yields, which would further hinder the model's accuracy as it is a multiple linear regression model.

4.2 Accuracy of predicting individual crops

Potato and Cassava have the two highest **RMSE** at [8.684] and [6.177] respectively. This means the model *might* be the least accurate in predicting the yields of these crops, and that the model does not capture the factors influencing Potato and Cassava yields as effectively as other crops, possibly due to their higher variability, unique crop-growing conditions, or missing predictors in the model.

In terms of the individual **RMSE**, while the absolute **RMSE** of potatoes is the highest [8.684], the mean yield of potatoes [18.204] is also the highest, resulting in the **ratio of RMSE to yield** [0.477] being the third best, thus it is not necessarily the most inaccurate at predicting the crop yield of potatoes. Instead, the model predicts that the yield of corn, which has a much lower average crop yield [3.66] and **RMSE** [2.547], but has the highest **ratio of RMSE to yield** [0.696], which is the most inaccurate. Some possibilities for this may be that the crop yield of corn may not be as affected by the chosen independent variables used in the model, and are instead dependent on other factors, and the presence of outliers which greatly affects the **RMSE**.

In conclusion, the accuracy of the models in predicting the crop yields in descending order is Soybean, Rice, Potato, Cassava, Wheat, then Corn.

4.3 Model Refinement

Due to the inherent nature of potatoes being the most resilient to harsh climate conditions, out of all the chosen staple crops, most of the time potatoes will be recommended as the ideal crop. To minimise the impact, we allow the user to freely pick certain selection crops to have a more useful selection of an ideal crop.

4.4 Limitations

- Crop yield does not account for the varying satiation or caloric contributions of different crops, e.g., rice absorbs water during cooking, making it more filling per gram.
- The model assumes linear relationships between the predictors and response variables, which oversimplifies real-world relationship.
- The model does not account for outliers contributed by extreme weather events that could disproportionately impact crop yields.

Conclusion

This analysis highlights the key factors influencing crop yields, including environmental and agricultural conditions. While the multiple linear regression model provides a rough estimate, it has limitations in capturing nutritional differences and other qualitative aspects of crops. Therefore, our model can guide farmers in optimising crop selection and resource allocation, but further research is needed for a holistic understanding to better address food security.

Appendix

Independent Variable	Coefficient (β)	Standard Error	t-statistic	p-value
Intercept	-4.066910151	1.337363879	-3.040989977	0.00239089
PM2.5	-0.027258809	0.009594756	-2.841011197	0.004545972
Annual Precipitation	0.000565777	0.000171246	3.303873423	0.000971659
Annual Mean Temperature	0.577419916	0.054306154	10.63267923	$1.12457E-25$
Pesticides per use area	0.181586962	0.038747069	4.686469645	$2.98E-06$

Table 1: Cassava

Independent Variable	Coefficient (β)	Standard Error	t-statistic	p-value
Intercept	8.239160227	0.146161926	56.37008533	0
PM2.5	-0.009019317	0.003149254	2.73	0.007
Annual Precipitation	-0.000579584	$5.46158E-05$	-10.61202503	$6.56017E-26$
Annual Mean Temperature	-0.217871897	0.006749173	-32.28127456	$6.0049E-200$
Pesticides per use area	0.280270438	0.012191543	22.98892273	$6.5072E-109$

Table 2: Corn

Independent Variable	Coefficient (β)	Standard Error	t-statistic	p-value
Intercept	27.54459282	0.434858017	63.3415776	0
PM2.5	-0.077277726	0.010894573	-7.093231055	$1.58633E-12$
Annual Precipitation	-0.003334066	0.000192622	-17.30884857	$2.12715E-64$
Annual Mean Temperature	-0.357678356	0.021826886	-16.38705372	$3.75957E-58$
Pesticides per use area	1.134804126	0.042363572	26.78726284	$1.3159E-143$

Table 3: Potato

Independent Variable	Coefficient (β)	Standard Error	t-statistic	p-value
Intercept	5.903122969	0.124416611	47.44642133	0
PM2.5	-0.007727598	0.002029587	-3.807472726	0.000143548
Annual Precipitation	-0.000251441	$3.65749E-05$	-6.874686003	$7.70284E-12$
Annual Mean Temperature	-0.103622285	0.005624481	-18.42343837	$1.86697E-71$
Pesticides per use area	0.166742396	0.008985052	18.5577555	$2.05108E-72$

Table 4: Rice

Independent Variable	Coefficient (β)	Standard Error	t-statistic	p-value
Intercept	2.315966822	0.047542112	48.71400768	0
PM2.5	-0.010988578	0.00115867	-9.483789474	$6.36979E-21$
Annual Precipitation	-0.000143069	$2.15949E-05$	-6.625138308	$4.3917E-11$
Annual Mean Temperature	-0.023249129	0.002278141	-10.20530612	$6.68942E-24$
Pesticides per use area	0.086370924	0.004994587	17.29290726	$9.34126E-63$

Table 5: Soybean

Independent Variable	Coefficient (β)	Standard Error	t-statistic	p-value
Intercept	4.588605057	0.093997323	48.81633778	0
PM2.5	−0.004473236	0.002399777	−1.864021745	0.062422345
Annual Precipitation	−0.000317733	$4.67021E-05$	−6.803401566	$1.2426E-11$
Annual Mean Temperature	−0.099603282	0.004840428	−20.57737275	$8.80861E-88$
Pesticides per use area	0.151961538	0.01010249	15.04198879	$2.99535E-49$

Table 6: Wheat

Peer Contribution

Member	Contributions
Megan	Coordinated tasks, compiled the report, contributed to Task 1 and Task 3.
Eddy	Processed data for Task 1, developed models for Task 2, supported Task 3.
Hoe Teng	Conducted literature reviews for Tasks 1 and 2, contributed to Task 3.
Zeyu	Built scripts for Task 2 models, handled sensitivity analysis for Task 3.
Edrea	Reviewed materials, created visualizations, managed presentation prep.